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|-------------|---------------|------------------|
| Surname     | Centre Number | Candidate Number |
| Other Names |               | 0                |

GCSE – **NEW**

3430U20-1



S18-3430U20-1

**SCIENCE (Double Award)****Unit 2: CHEMISTRY 1  
FOUNDATION TIER**

WEDNESDAY, 13 JUNE 2018 – MORNING

1 hour 15 minutes

**ADDITIONAL MATERIALS**

In addition to this examination paper you will need a calculator and a ruler.

**INSTRUCTIONS TO CANDIDATES**

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

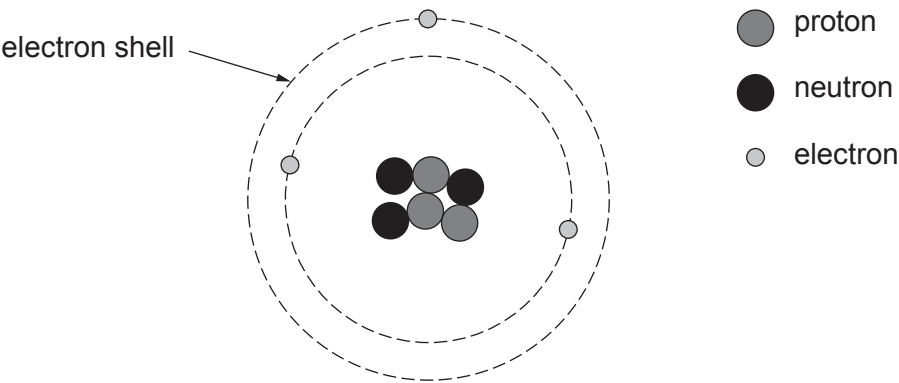
| For Examiner's use only |              |              |
|-------------------------|--------------|--------------|
| Question                | Maximum Mark | Mark Awarded |
| 1.                      | 7            |              |
| 2.                      | 9            |              |
| 3.                      | 7            |              |
| 4.                      | 6            |              |
| 5.                      | 10           |              |
| 6.                      | 6            |              |
| 7.                      | 8            |              |
| 8.                      | 7            |              |
| <b>Total</b>            | <b>60</b>    |              |



JUN183430U20101

Answer all questions.

1. (a) The diagram represents a lithium atom. Its nucleus contains protons and neutrons.



(i) Complete the table of information about particles in the atom. [2]

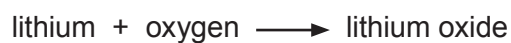
| Name of particle | Charge | Mass        |
|------------------|--------|-------------|
| proton           |        | 1           |
| neutron          | 0      |             |
| electron         | -1     | almost zero |

(ii) Use the information in the diagram to give the mass number of this lithium atom. [1]

.....



(b) Lithium reacts with oxygen. The word equation for this reaction is as follows.



(i) Name the **metal** in this equation. [1]

.....

(ii) Name the **compound** in this equation. [1]

.....

(iii) Name a **reactant** in this equation. [1]

.....

(iv) Lithium oxide contains the ions  $\text{Li}^+$  and  $\text{O}^{2-}$ .

Underline the formula of lithium oxide. [1]



2. Peter, Claire and David compared the hardness of three samples of water, **A**, **B** and **C**. They added soap solution to each sample until it gave a lather on shaking. They carried out a fair test.

(a) Give **two** ways they made it a fair test. [2]

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.....

(b) Their results are shown in the table.

| Water sample | Volume of soap solution needed to give a lather (cm <sup>3</sup> ) |        |       |       |
|--------------|--------------------------------------------------------------------|--------|-------|-------|
|              | Peter                                                              | Claire | David | Mean  |
| <b>A</b>     | 18                                                                 | 19     | 17    | ..... |
| <b>B</b>     | 8                                                                  | 10     | 9     | 9     |
| <b>C</b>     | 1                                                                  | 1      | 1     | 1     |

- (i) Calculate the mean volume of soap solution needed for sample **A** to give a lather. **Show your working.** [2]

Mean = ..... cm<sup>3</sup>

- (ii) Which of the three water samples is the **hardest**? Give a reason for your answer. [2]

.....

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(c) The box shows some advantages and some disadvantages of hard water.

|                                    |                                    |
|------------------------------------|------------------------------------|
| <b>strengthens bones and teeth</b> | <b>forms limescale when heated</b> |
| <b>wastes soap</b>                 | <b>reduces heart disease</b>       |
| <b>forms a scum with soap</b>      | <b>tastes better</b>               |

Choose **all** the disadvantages of hard water from the box.

[2]

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(d) Choose from the box an ion that causes hardness in water.

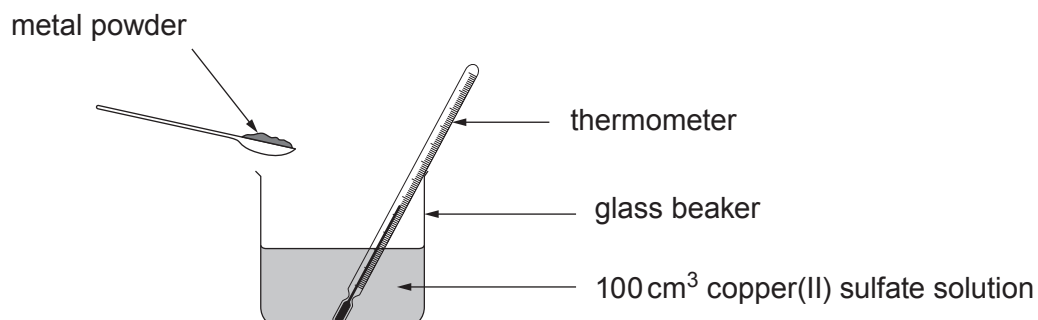
[1]

|                      |                       |                        |                        |
|----------------------|-----------------------|------------------------|------------------------|
| <b>K<sup>+</sup></b> | <b>Na<sup>+</sup></b> | <b>Ca<sup>2+</sup></b> | <b>Cu<sup>2+</sup></b> |
|----------------------|-----------------------|------------------------|------------------------|

Ion .....



3. Magnesium, zinc and iron powders were each added separately to 100 cm<sup>3</sup> of copper(II) sulfate solution, to see which gave the greatest temperature change.



The temperature was recorded before and after each reaction. The results are shown in the table.

| Metal     | Temperature before the reaction (°C) | Temperature after the reaction (°C) | Temperature increase (°C) |
|-----------|--------------------------------------|-------------------------------------|---------------------------|
| zinc      | 20                                   | .....                               | 14                        |
| magnesium | 19                                   | 39                                  | 20                        |
| iron      | 19                                   | 24                                  | 5                         |

- (a) Calculate the temperature after the reaction with zinc.

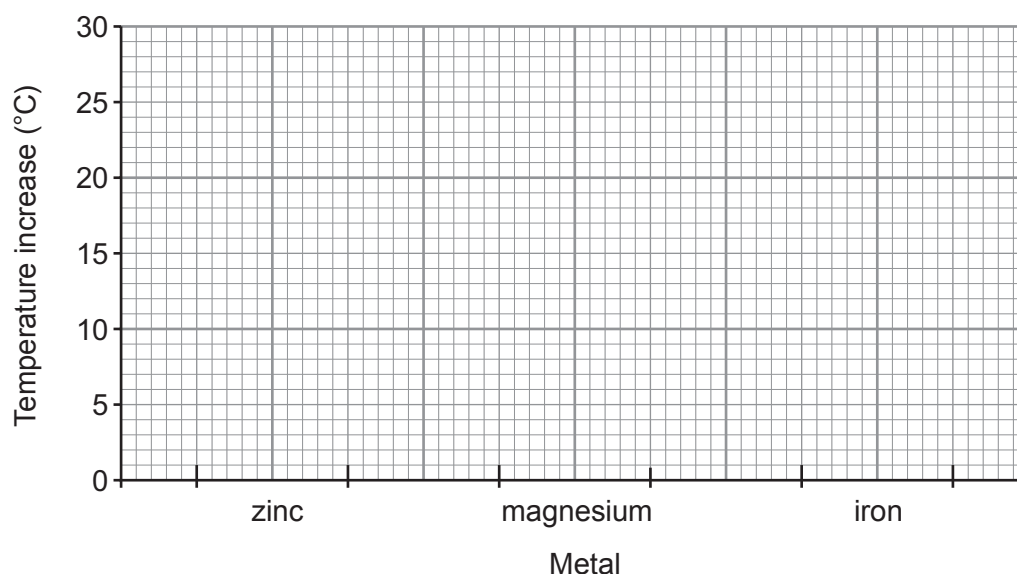
[1]

Temperature = ..... °C



- (b) Plot a **bar chart** to show the temperature increase for each metal.

[2]



- (c) (i) Calculate the energy released in the reaction with magnesium. Use the following equation.

[2]

$$\begin{array}{ccccc} \text{energy released} & = & \text{volume of solution} & \times & 4.2 & \times & \text{temperature increase} \\ \text{(J)} & & \text{(cm}^3\text{)} & & & & \text{(}^\circ\text{C)} \end{array}$$

Energy released = ..... J

- (ii) In this reaction magnesium sulfate,  $\text{MgSO}_4$ , is formed. What is the relative formula mass ( $M_r$ ) of magnesium sulfate?

[2]

$$A_r(\text{Mg}) = 24 \quad A_r(\text{S}) = 32 \quad A_r(\text{O}) = 16$$

$M_r$  = .....



4. A student investigated changes that happened when pairs of colourless solutions were mixed. The table shows the observations made after mixing.

| Pair of solutions | Appearance of the reactants | Appearance when mixed   | Temperature change when mixed |
|-------------------|-----------------------------|-------------------------|-------------------------------|
| <b>A</b>          | two colourless solutions    | no change               | increase                      |
| <b>B</b>          | two colourless solutions    | bubbles form            | increase                      |
| <b>C</b>          | two colourless solutions    | white precipitate forms | no change                     |
| <b>D</b>          | two colourless solutions    | no change               | no change                     |

- (a) Which pair of solutions do **not** react? Give a reason for your answer. [2]

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- (b) (i) The gas given off when pair **B** react is carbon dioxide. Describe the test for carbon dioxide gas. [1]

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.....

- (ii) One of the solutions in pair **B** is sodium carbonate. Give the formula of sodium carbonate. [1]

.....

- (c) (i) Which pair of solutions is silver nitrate and sodium chloride? [1]

.....

- (ii) What colour would sodium chloride give in a flame test? [1]

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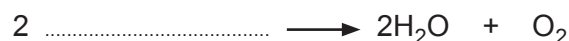
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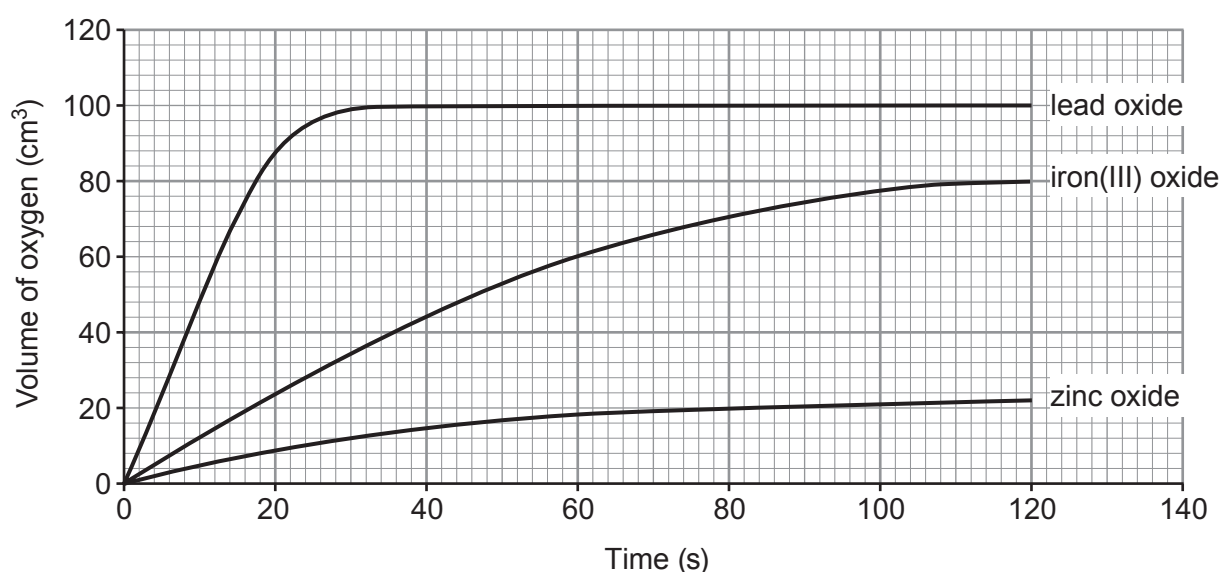


5. Hydrogen peroxide solution decomposes to give water and oxygen gas. Many metal oxides act as catalysts for the reaction.

(a) Use the symbol equation for the reaction to find the formula of hydrogen peroxide. [1]



(b) Catalysts are used to speed up the reaction. The graphs show the volume of oxygen formed over 120 seconds when 1.0g of different metal oxides were added to 50 cm<sup>3</sup> of hydrogen peroxide solution.



- (i) Use the graph to find the time taken to form 44 cm<sup>3</sup> of oxygen using iron(III) oxide as the catalyst. [1]

..... s



- (ii) State which metal oxide is the **best** catalyst. Give a reason for your answer. [1]

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- (iii) After 120 seconds, the contents of each beaker were washed into a filter paper and funnel. The catalyst left over on each filter paper was dried and weighed.

Tick (✓) the box next to the statement which is correct. [1]

the same mass of all catalysts is left over

☐

more zinc oxide is left over than lead oxide

☐

about 80 % of the iron(III) oxide is left over

☐

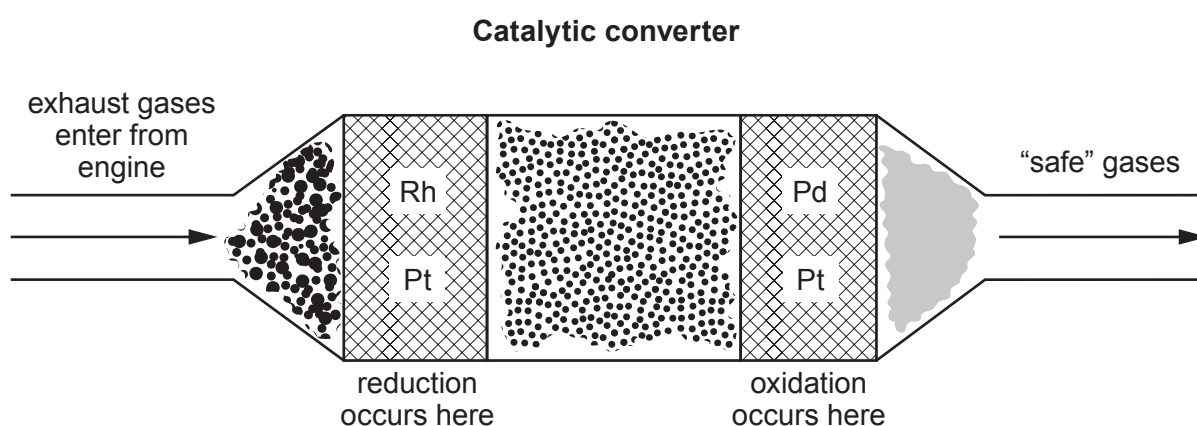
no lead oxide is left over

☐

- (c) Exhaust gases from car engines contain harmful molecules, for example, carbon monoxide and nitrogen oxides. All cars are fitted with catalytic converters that split up these harmful molecules.

The catalysts are made from platinum (Pt) and palladium (Pd) or rhodium (Rh). A mesh structure is used that exposes the maximum surface area of catalyst to the exhaust gases, while also reducing the amount of catalyst required. Platinum, palladium and rhodium are extremely expensive.

As the exhaust gases from the engine pass over the catalysts, chemical reactions take place on their surfaces. The harmful molecules are broken up and converted into other gases that are “safe” to enter the air. These gases include carbon dioxide, nitrogen, oxygen and water.



There are two different types of catalyst, a **reduction catalyst** and an **oxidation catalyst**. The reduction catalyst removes oxygen from nitrogen oxides to make nitrogen and oxygen. The oxidation catalyst adds oxygen to carbon monoxide to make carbon dioxide.

One of the biggest drawbacks of the catalytic converter is that it only works effectively at high temperatures. The average running temperature of a car engine is between 90 and 110°C. It takes nearly 30 minutes for these temperatures to be reached.

The table opposite shows the percentages of carbon monoxide and nitrogen oxides converted to safe gases by a catalytic converter at different temperatures.



| Temperature (°C) | Carbon monoxide converted (%) | Nitrogen oxides converted (%) |
|------------------|-------------------------------|-------------------------------|
| 25               | 16                            | 25                            |
| 50               | 19                            | 28                            |
| 75               | 26                            | 35                            |
| 100              | 60                            | 72                            |
| 125              | 91                            | 92                            |
| 150              | 93                            | 94                            |
| 175              | 95                            | 95                            |
| 200              | 97                            | 98                            |

- (i) Using your knowledge of particle theory suggest why a catalyst in mesh form works better than a lump of catalyst. [2]

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.....

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- (ii) Tick (✓) the box that best describes the adverse effect that gases leaving the exhaust would have on the environment. [1]

they have no effect on the environment

☐

they reduce oxygen levels

☐

they deplete the ozone layer

☐

they cause global warming

☐


- (iii) Tick (✓) the box that best describes the conversion of carbon monoxide and nitrogen oxides into "safe" gases at different temperatures. [1]

equal amounts of carbon monoxide and nitrogen oxides  
are converted at every temperature

☐

more carbon monoxide is converted than nitrogen oxides  
up to 100 °C

☐

more nitrogen oxides are converted than carbon monoxide  
up to 100 °C

☐

40 % more nitrogen oxides are converted than carbon monoxide  
up to 100 °C

☐

- (iv) In your opinion how effective are catalytic converters? Explain your answer. [2]

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Examiner  
only

6. Describe how respiration and photosynthesis keep the carbon dioxide and oxygen content of the atmosphere approximately constant. Discuss how human activity is threatening this balance.  
[6 QER]

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| 6 |



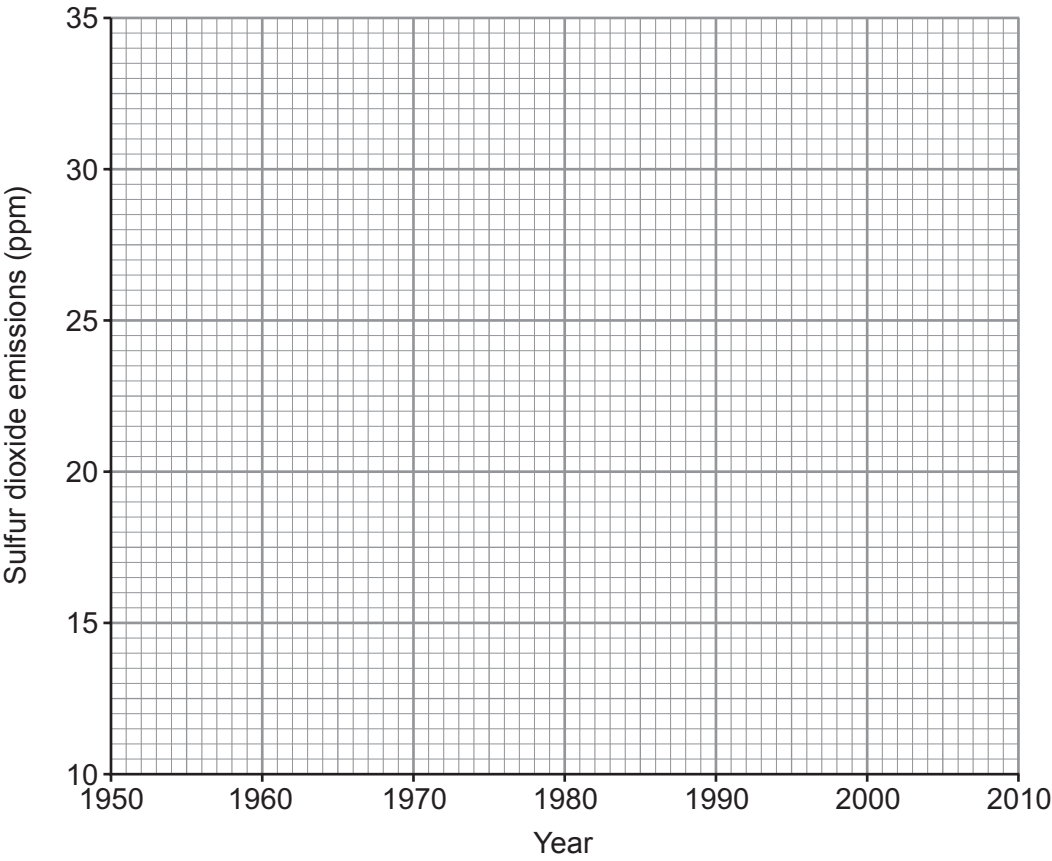
7. Burning fossil fuels containing sulfur causes sulfur dioxide, SO<sub>2</sub>, to be released into the atmosphere.

The table shows sulfur dioxide emissions in the UK between 1950 and 2010.

| Year | Sulfur dioxide emissions (ppm) |
|------|--------------------------------|
| 1950 | 12.0                           |
| 1960 | 16.0                           |
| 1970 | 21.5                           |
| 1980 | 29.5                           |
| 1990 | 29.0                           |
| 2000 | 24.0                           |
| 2010 | 18.5                           |

ppm = parts per million

- (a) (i) On the grid plot the sulfur dioxide emissions against the year and draw a suitable line. [3]





- (ii) Describe how sulfur dioxide emissions changed between 1950 and 2010. [2]

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.....

.....

- (iii) The UK government introduced a regulation to reduce sulfur dioxide emissions in the 1980s. From your graph, state why it is difficult to decide exactly the year when the regulation came into force. [1]

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.....

.....

- (b) Sulfur dioxide can be converted to sulfur and water by reacting it with hydrogen sulfide, H<sub>2</sub>S.

Complete and balance the symbol equation for this reaction. [2]



8. The table shows some physical properties of Group 6 elements.

| Element   | Melting point (°C) | Boiling point (°C) | Density (g/cm <sup>3</sup> ) | Electrical conductor |
|-----------|--------------------|--------------------|------------------------------|----------------------|
| oxygen    | −219               | −183               | 0.0014                       | no                   |
| sulfur    | 115                | 445                | 2.0                          | no                   |
| selenium  | 221                | 685                | 4.8                          | semi-conductor       |
| tellurium | 450                | 988                | 6.2                          | semi-conductor       |

(a) (i) Describe the trend in the melting points of the Group 6 elements. [1]

.....

(ii) Give the physical state of selenium at 400 °C. Give a reason for your choice. [2]

.....

.....

(iii) Explain why it is difficult to classify selenium as either a metal or a non-metal. [1]

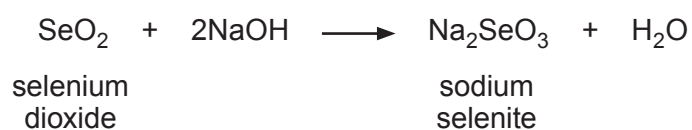
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(b) Selenium dioxide reacts with sodium hydroxide to produce sodium selenite and water.



(i) Calculate the relative formula mass ( $M_r$ ) of sodium selenite,  $\text{Na}_2\text{SeO}_3$ . [1]

$$A_r(\text{Na}) = 23 \qquad A_r(\text{Se}) = 79 \qquad A_r(\text{O}) = 16$$

$M_r =$  .....

(ii) Calculate the percentage by mass of selenium in sodium selenite. [2]

Percentage = ..... %

|   |
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|   |
| 7 |

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### FORMULAE FOR SOME COMMON IONS

| POSITIVE IONS |                  | NEGATIVE IONS |                    |
|---------------|------------------|---------------|--------------------|
| Name          | Formula          | Name          | Formula            |
| aluminium     | $\text{Al}^{3+}$ | bromide       | $\text{Br}^-$      |
| ammonium      | $\text{NH}_4^+$  | carbonate     | $\text{CO}_3^{2-}$ |
| barium        | $\text{Ba}^{2+}$ | chloride      | $\text{Cl}^-$      |
| calcium       | $\text{Ca}^{2+}$ | fluoride      | $\text{F}^-$       |
| copper(II)    | $\text{Cu}^{2+}$ | hydroxide     | $\text{OH}^-$      |
| hydrogen      | $\text{H}^+$     | iodide        | $\text{I}^-$       |
| iron(II)      | $\text{Fe}^{2+}$ | nitrate       | $\text{NO}_3^-$    |
| iron(III)     | $\text{Fe}^{3+}$ | oxide         | $\text{O}^{2-}$    |
| lithium       | $\text{Li}^+$    | sulfate       | $\text{SO}_4^{2-}$ |
| magnesium     | $\text{Mg}^{2+}$ |               |                    |
| nickel        | $\text{Ni}^{2+}$ |               |                    |
| potassium     | $\text{K}^+$     |               |                    |
| silver        | $\text{Ag}^+$    |               |                    |
| sodium        | $\text{Na}^+$    |               |                    |
| zinc          | $\text{Zn}^{2+}$ |               |                    |



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



24

|                                    |
|------------------------------------|
| <sup>1</sup><br>H<br>Hydrogen<br>1 |
|------------------------------------|

|                                        |                                        |
|----------------------------------------|----------------------------------------|
| <sup>7</sup><br>Li<br>Lithium<br>3     | <sup>9</sup><br>Be<br>Beryllium<br>4   |
| <sup>23</sup><br>Na<br>Sodium<br>11    | <sup>24</sup><br>Mg<br>Magnesium<br>12 |
| <sup>39</sup><br>K<br>Potassium<br>19  | <sup>40</sup><br>Ca<br>Calcium<br>20   |
| <sup>86</sup><br>Rb<br>Rubidium<br>37  | <sup>88</sup><br>Sr<br>Strontium<br>38 |
| <sup>133</sup><br>Cs<br>Caesium<br>55  | <sup>137</sup><br>Ba<br>Barium<br>56   |
| <sup>223</sup><br>Fr<br>Francium<br>87 | <sup>226</sup><br>Ra<br>Radium<br>88   |
|                                        | <sup>227</sup><br>Ac<br>Actinium<br>89 |

|                                        |                                        |                                        |                                         |                                         |                                      |
|----------------------------------------|----------------------------------------|----------------------------------------|-----------------------------------------|-----------------------------------------|--------------------------------------|
| <sup>11</sup><br>B<br>Boron<br>5       | <sup>12</sup><br>C<br>Carbon<br>6      | <sup>14</sup><br>N<br>Nitrogen<br>7    | <sup>16</sup><br>O<br>Oxygen<br>8       | <sup>19</sup><br>F<br>Fluorine<br>9     | <sup>20</sup><br>Ne<br>Neon<br>10    |
| <sup>27</sup><br>Al<br>Aluminium<br>13 | <sup>28</sup><br>Si<br>Silicon<br>14   | <sup>31</sup><br>P<br>Phosphorus<br>15 | <sup>32</sup><br>S<br>Sulfur<br>16      | <sup>35.5</sup><br>Cl<br>Chlorine<br>17 | <sup>40</sup><br>Ar<br>Argon<br>18   |
| <sup>70</sup><br>Ga<br>Gallium<br>31   | <sup>73</sup><br>Ge<br>Germanium<br>32 | <sup>75</sup><br>As<br>Arsenic<br>33   | <sup>79</sup><br>Se<br>Selenium<br>34   | <sup>80</sup><br>Br<br>Bromine<br>35    | <sup>84</sup><br>Kr<br>Krypton<br>36 |
| <sup>115</sup><br>In<br>Indium<br>49   | <sup>119</sup><br>Sn<br>Tin<br>50      | <sup>122</sup><br>Sb<br>Antimony<br>51 | <sup>128</sup><br>Te<br>Tellurium<br>52 | <sup>127</sup><br>I<br>Iodine<br>53     | <sup>131</sup><br>Xe<br>Xenon<br>54  |
| <sup>204</sup><br>Tl<br>Thallium<br>81 | <sup>207</sup><br>Pb<br>Lead<br>82     | <sup>209</sup><br>Bi<br>Bismuth<br>83  | <sup>210</sup><br>Po<br>Polonium<br>84  | <sup>210</sup><br>At<br>Astatine<br>85  | <sup>222</sup><br>Rn<br>Radon<br>86  |

24

Key

|                                                 |                      |               |
|-------------------------------------------------|----------------------|---------------|
| <sup>A<sub>r</sub></sup><br>Symbol<br>Name<br>Z | relative atomic mass | atomic number |
|-------------------------------------------------|----------------------|---------------|